VAILS: A Voice Agent for Independent Living Support

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Abstract. Maintaining independence at home poses significant challenges for older adults and those with cognitive impairment, particularly in medication adherence, daily wellness monitoring, and access to simple everyday assistance. This paper presents VAILS (Voice Agent for Independent Living Support), a conversational AI-powered collaborator that integrates medication management, proactive wellness monitoring, and on-demand conversational assistance through a unified voice interface. VAILS delivers scheduled medication reminders with confirmation prompts, automatically logging adherence, and triggers caregiver notifications when medication or wellness escalation conditions are met. To improve adaptability and decision-making, VAILS incorporates three machine learning models: a supervised adherence forecast, a supervised risk classifier, and unsupervised behavioural clustering, whose outputs are presented as interpretable advisory signals for caregivers. Beyond laboratory feasibility tests, we report a preliminary seven-day in-home pilot with one older adult, during which VAILS delivered 43 scheduled events, logged an overall medication adherence of 80.6%, and per-dose logging isolated a single problematic evening dose. This work demonstrates the end-to-end feasibility of intelligent voice agents for assisted living and provides a template for larger field evaluation.

Keywords: Voice Agents, Independent Living, Medication Adherence, Wellness Monitoring, Elderly Care, Conversational AI, Machine Learning.

1 Introduction

Europe is ageing rapidly: in 2024, 21.6% of the European Union (EU) population — some 97 million people — was aged 65 or over [1], while worldwide the number of people aged 60 and over is expected to double by 2050, reaching 2.1 billion [2]. Cognitive difficulties are common: about 15% of people in the EU (around 67 million) report difficulties with memory or concentration [3]. Moreover, roughly one in three older people in the EU lives alone (40.2% of women and 21.8% of men aged 65 or over) [4], often without daily support in managing their health routines. Medication management is a central point of failure: about half of chronic patients do not take their medication as prescribed, and non-adherence accounts for approximately 10% of all hospital admissions among older adults and is associated with accelerated disease progression and markedly increased healthcare costs [5].

Technological interventions have long been proposed to bridge the gap between prescribed care regimens and actual patient behaviour. However, mainstream solutions remain predominantly screen- and touch-centric, poorly suited to individuals with limited technological literacy, reduced fine motor control, or impaired vision. Voice interaction reduces reliance on screens and touch while leveraging the familiar modality of spoken language.

Recent advances in large language models (LLMs) and streaming voice APIs have opened a new design space: voice agents capable of natural, contextually adaptive dialogue rather than rigid command-and-response exchanges. At the same time, machine learning (ML) over longitudinal behavioural data enables systems that not only respond to users but anticipate their needs and adapt to their evolving patterns.

Despite these advances, existing systems rarely combine voice-first interaction, medication confirmation, wellness check-ins, caregiver escalation, on-demand conversational assistance, and adaptive longitudinal risk modelling in a single lightweight architecture. In this paper, we introduce VAILS (Voice Agent for Independent Living Support), an implemented system prototype designed to operationalise these capabilities in a practical form. VAILS addresses three interrelated needs of its target population: a) reliable medication management with adherence tracking and caregiver escalation; b) proactive wellness monitoring with structured escalation; and c) accessible conversational support for everyday independent living. Conversational interaction is used across the system for medication follow-up, wellness check-ins, everyday current-information questions, and user-defined reminders. For current-information queries (e.g. time, weather, calendar questions, religious holidays, recent sports results), VAILS uses the Google Search grounding capability of the Gemini Live API [6, 7], while a dedicated reminder tool interprets natural-language ad hoc reminder requests and converts them into scheduled reminder events — all within a voice-first interface powered by the Gemini Live API, a real-time conversational interface built on top of a large language model.

The primary contributions of this work are:

- An integrated voice-first architecture that combines medication management, wellness monitoring, and conversational assistance, including current-information queries and natural-language ad hoc reminders, within a single system;
- A structured caregiver escalation workflow triggered by non-response, with escalation urgency informed by ML-based risk classification;
- An ML-based analytics layer that uses longitudinal interaction history for adherence forecasting, risk classification, and behavioural clustering, with explicit guidance on how caregivers should interpret each model output;
- A working prototype and functional feasibility demonstration covering the complete reminder, follow-up, wellness, escalation, and notification pipeline;
- A preliminary seven-day in-home pilot with one older adult, providing first evidence of end-to-end operation in a real home and illustrating how the ML analytics translate interaction logs into caregiver guidance.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 presents the problem and system requirements. Section 4 describes the

VAILS architecture and prototype. Section 5 presents the evaluation, combining feasibility tests with a preliminary in-home pilot, and Section 6 concludes with limitations and future work.

2 Related Work

2.1 Medication Adherence Technologies

Automated medication reminders have been studied extensively in the mHealth literature. Dayer et al. [8] showed that smartphone-based adherence applications rely largely on reminder notifications, manual entry, and app-based logging, remaining dependent on sustained user engagement, while Sullivan et al. [9] found that adherence feedback can motivate self-management changes in older adults with cognitive impairment, suggesting that alert-only systems provide limited support. Platforms such as Medisafe [10] add caregiver-oriented monitoring, but still rely primarily on screen-based interaction.

Voice-based medication reminders using smart speakers have also been explored. Shade et al. [11] found that Google Assistant reminders for scheduled pain medication can cue medication-taking behaviour in older adults, but with substantial variability in the timing of medication use after the reminder. These findings motivate reminder workflows that go beyond one-way prompting by supporting spoken confirmation, adherence logging, and caregiver escalation.

2.2 Wellness Monitoring and Escalation

Ambient assisted living (AAL) systems have commonly monitored daily routines through motion sensors, wearables, and smart-home infrastructure [12]. Such systems support early detection of deviations, but depend on installed hardware and may raise acceptability and privacy concerns in domestic settings, motivating complementary active-monitoring approaches that rely on scheduled check-ins rather than continuous passive sensing.

Escalation in many AAL deployments is implemented through rule- or threshold-based alerting, for example notifying caregivers when activity patterns suggest illness or safety concerns [13]. Repeated low-specificity alerts, however, contribute to alarm fatigue, a documented patient safety concern [14]. Risk-informed escalation distinguishes routine delayed responses from genuinely anomalous non-response patterns, allowing caregiver notifications to be prioritised appropriately.

2.3 Conversational AI for Ageing Populations

Kim [15] found that older adults valued the hands-free and eyes-free affordances of smart-speaker assistants during first use, while Trajkova and Martin-Hammond [16] showed that long-term adoption can be limited by unclear value, difficulty discovering useful functions, and challenges in shared spaces. Voice interaction alone is therefore

not sufficient: systems must support flexible phrasing, graceful error recovery, and clearly meaningful everyday tasks.

In healthcare, conversational agents have been studied for education, behaviour change, and patient support, with systematic reviews identifying both promise and limitations [17], and generative voice agents are expected to further expand this role [18]. Deployed systems such as ElliQ illustrate the potential of proactive conversational companions that combine everyday conversation, reminders, wellness prompts, and social engagement [19]. VAILS builds on this trajectory by combining structured assisted-living workflows with open-ended conversational support, including everyday current-information questions answered through Google Search grounding [6, 7] and natural-language creation of ad hoc reminders.

2.4 Machine Learning in Eldercare

Machine learning has been widely applied in eldercare to support fall detection, activity recognition, and anomaly detection from sensors or smart-home data [12, 20, 21]. These approaches demonstrate the value of modelling routine behaviour over time, but typically depend on passive sensing infrastructure or wearables; a complementary direction is to model the interaction traces generated by voice reminders and wellness check-ins.

Medication-adherence research has also explored ML models for predicting future non-adherence from prior adherence, clinical, questionnaire, claims, or behavioural data [21]. In the system presented here, supervised models estimate the likelihood of timely intake and classify escalation risk, while unsupervised clustering groups similar response patterns over time. The goal is not to diagnose health status, but to support caregiver prioritisation while keeping the notification workflow auditable.

2.5 Summary and Research Gap

Overall, prior work demonstrates the value of digital reminders, voice interfaces, ambient monitoring, and ML-based behavioural modelling for assisted living, but these strands are often developed separately: medication tools remain screen-based or reminder-oriented, voice assistants rarely support structured adherence confirmation and escalation, and ML approaches frequently depend on external clinical or sensor datasets. VAILS addresses this gap by integrating all of them into a single lightweight prototype.

3 Problem Description and Motivation

Supporting independent living requires more than issuing reminders: the core challenge is to coordinate medication adherence, wellbeing awareness, and caregiver involvement in ways that fit everyday routines and remain usable for people with limited memory, vision, dexterity, or confidence with digital devices. This section derives the requirements that guide VAILS from three connected concerns: medication non-adherence, caregiver burden, and the interaction limits of existing tools.

3.1 Medication Non-Adherence and Its Consequences

Medication non-adherence is among the most persistent challenges in the long-term management of chronic conditions. Osterberg and Blaschke [22] estimate that approximately 50% of patients with chronic diseases fail to take medications as prescribed. Cutler et al. [23] report that medication non-adherence is associated with annual per-person economic losses ranging from $949 to $44,190 across disease groups, primarily due to avoidable hospitalisations and disease progression. For older adults, these risks are compounded by complex regimens and multiple daily schedules.

Cognitive impairment further increases this risk: dementia directly impairs reliable medication management, with forgotten doses and confused schedules among the most common barriers [5], and even mild cognitive impairment can reduce the capacity for independent self-management, often requiring caregiver involvement before a formal diagnosis [24].

3.2 Caregiver Burden

When independent medication management becomes unreliable, monitoring responsibility often shifts to informal caregivers, imposing substantial psychological and practical strain: caregiving burden is strongly associated with anxiety ($r = 0.51$) [25], and caregiving intensity with perceived burden across chronic disease contexts [26].

These findings imply that assisted living systems should reduce routine monitoring work without hiding clinically relevant events; excessive or poorly calibrated alerts risk alarm fatigue [14]. A useful system therefore needs escalation logic that is visible, conservative, and sensitive to the user's observed pattern of response.

3.3 Limitations of Existing Approaches

Existing technological interventions address parts of this problem but leave important gaps. Screen-based reminder applications require active user interaction, often unsuitable for older adults with reduced motor control, impaired vision, or limited digital literacy [27], and remain fundamentally passive: they issue reminders but cannot verify intake, interpret non-response, or adapt escalation to individual behaviour.

Voice assistants reduce interaction burden [11, 15], yet current commercial systems primarily deliver one-way notifications, lacking structured conversational follow-up, confirmation logging, and the ability to distinguish a delayed response from a potentially concerning non-response; as a result, they often fail to integrate into everyday care routines [16].

Conversational interfaces also add a social dimension relevant for people living alone: brief, natural exchanges create a sense of presence that may support long-term acceptance, making natural language both an accessibility mechanism and a means of sustaining engagement.

3.4 System Requirements

These challenges motivate explicit functional and non-functional requirements. Functionally, the system must deliver scheduled medication reminders through natural speech; support spoken confirmation of intake through conversational follow-up; notify caregivers when the user does not respond or reports a concern; calibrate escalation urgency using the user's observed behavioural risk level; support proactive wellness monitoring; provide on-demand conversational assistance for simple everyday questions; allow users to create ad hoc reminders through natural language; and provide a caregiver dashboard for scheduling, monitoring, and analytics.

Non-functional requirements are equally important: the senior should not need to operate a screen or smartphone during routine use, and the system should run on standard consumer hardware without specialised sensing infrastructure, store sensitive data locally where possible, and provide robust fallbacks when voice interaction or network services fail.

These requirements define the design target for VAILS: a low-friction voice interface for the senior user combined with structured logging, adaptive risk analytics, and caregiver-facing oversight. The next section describes how VAILS implements this design.

4 The VAILS System

VAILS is implemented as a lightweight two-process Python system that separates assisted-living logic from caregiver interaction. A FastAPI backend mediates all database access, scheduling, voice sessions, notification dispatch, and analytics, while a Streamlit dashboard communicates with the backend through authenticated REST calls. This separation keeps database writes behind the API boundary, allows the dashboard to run on a separate host if needed, and makes the system easier to deploy on commodity hardware. Figure 1 summarises the architecture.

4.1 Medication and Wellness Workflows

The medication management subsystem stores reminder schedules and intake logs in SQLite. At startup, scheduled reminders are loaded into APScheduler using cron-style triggers. Schedules support daily, weekly, monthly, and one-time repetition patterns. When a medication reminder fires, VAILS generates a personalised spoken prompt and plays it through a layered text-to-speech pipeline. The primary synthesiser is attempted first, followed by network-based and platform-level fallbacks, so that a reminder can still be delivered when a preferred audio path is unavailable. General reminders can be logged as played, whereas medication reminders initiate a follow-up dialogue after a short configurable delay. TTS delivery respects per-senior voice preferences (volume and speech rate), accommodating hearing difficulties or individual comfort, and wellness check-ins are bounded by a caregiver-defined active time window (e.g. 09:00–21:00) so that automated prompts are not delivered at night.

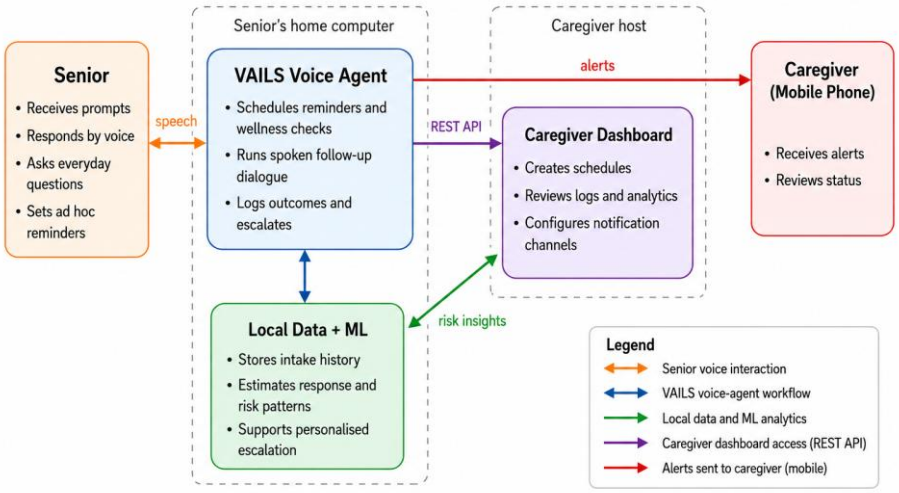


Fig. 1. VAILS high-level architecture. The senior interacts through a voice-first agent, while local data and ML analytics support caregiver dashboard review, risk-aware escalation, and notification workflows.

The follow-up dialogue asks the senior whether the medication was taken and records the result as taken, skipped, missed, or unanswered, converting the reminder from a one-way notification into a monitored adherence event. The same pattern extends to wellness monitoring: caregiver-defined check-ins ask short questions, classify the response as normal or concerning, and trigger escalation on a reported problem or non-response. Both workflows share the same design principle: routine events are logged quietly, while non-response or concerning responses are made visible to caregivers.

4.2 Voice Interaction Layer

The voice interface is built around real-time spoken interaction with the Gemini Live API [6]. Reminder-context sessions are initiated by the scheduler and use structured prompts that instruct the model to end the dialogue with machine-parseable status markers, such as whether the medication was taken, skipped, or missed; the markers are written to the intake log, keeping the workflow auditable even though the interaction itself is conversational.

VAILS also supports hands-free ambient assistance through voice activation. An energy-based voice activity detector (VAD) monitors microphone input and, when speech is detected, a wake-word check with Greek and English phonetic variants is applied to a short segment of buffered audio, reducing unintended activations from background speech or television audio; a session lock prevents overlapping conversations. Interactions are configured primarily for Greek speech, with per-user language preferences, and location-aware Gemini Live sessions with Google Search grounding support everyday current-information questions. User-defined reminder requests are handled

through native function calling: the model extracts the reminder content and target time from the utterance and invokes a registered set reminder tool, so seniors can create short-term ad hoc reminders without using a screen.

4.3 ML-Based Risk Analytics

The analytics layer is based on interaction history. Its purpose is not diagnosis, but caregiver prioritisation: the models transform reminder and wellness logs into interpretable signals about response likelihood, adherence stability, and escalation risk, allowing the dashboard to present risk-aware recommendations without requiring clinical labels at deployment time.

Three complementary models are used, all implemented with scikit-learn [28] and trained on demand per senior, using only that individual's own interaction history stored in the database; no model state is persisted between sessions. Since no adequate real-world dataset from field evaluations in real homes is yet available, the models were trained with synthetic data: a generator populated the database with 60 days of realistic interaction logs, simulating response delays, missed reminders, and adherence variability across morning, afternoon, and evening slots. In a real deployment the same training logic operates on accumulated real interaction logs, as demonstrated in the pilot of Section 5. Table 1 summarises the three models and how caregivers should read their outputs.

Table 1. The three ML models of the VAILS analytics layer and the interpretation of their outputs by the caregiver or supervising physician.

Model	Algorithm	Output	Interpretation for the caregiver
Adherence forecast	Logistic regression (supervised)	Intake probability (0–100%) per dose slot	Likelihood of timely intake per slot; low values (e.g. < 70%) flag slots needing a proactive check or schedule change
Risk classification	Decision tree (supervised)	Risk class: Low / Medium / High	Escalation urgency: Low = routine monitoring; Medium = follow-up check recommended; High = immediate caregiver alert
Behaviour clustering	k-means, k = 3 (unsupervised)	Weekly adherence profile (cluster assignment)	Positions current week against the senior's own history; drift toward a higher-miss profile signals a deteriorating routine

First, an adherence forecast model uses logistic regression to estimate the probability that the senior will confirm intake of a scheduled dose. Its features are the hour of day, the day of week, and the recent taken ratio computed over the last seven doses; the binary training target is whether a past dose was confirmed as taken. Second, a risk classification model uses a shallow decision tree over rolling 30-day behavioural features — missed-dose count, mean response delay in minutes, and trailing consecutive-

miss streak — and assigns one of three classes: Low, Medium, or High. Training labels are bootstrapped from the auditable rule-based risk score of Table 2, so the classifier learns a generalisable approximation of the rule set. Third, a behavioural clustering model applies k-means ($k = 3$) over weekly windows described by missed count and mean response delay, grouping weeks into adherence profiles that range from stable to irregular.

To preserve interpretability, the ML outputs are combined with rule-based pattern detection: recent missed doses, repeated non-response, evening misses, and consecutive misses are surfaced as caregiver-readable signals rather than opaque outputs.

The three models are deliberately positioned as an advisory tool for the caregiver or supervising physician, not as an autonomous decision maker: escalation in the current prototype remains driven by the transparent rule-based risk score, and the ML outputs contextualise and prioritise what the caregiver sees on the dashboard. Once validated in a field evaluation yielding a high-quality dataset from many older adults and caregivers, ML-driven escalation could progressively replace the rule-based score in the notification workflow.

4.4 Caregiver Dashboard, Escalation, and Access Control

The caregiver dashboard provides scheduling, monitoring, analytics, and notification configuration through a browser interface. Caregivers can create medication and wellness schedules, inspect intake logs, review adherence trends, and view ML-derived risk indicators. The dashboard communicates only through the backend API, enforcing the same access-control rules as other clients.

Escalation follows a graduated protocol. A confirmed or inferred missed dose triggers Telegram and/or SMS notifications according to the configured contact settings; an unanswered wellness check first schedules a retry and then, if the senior remains unresponsive, sends a no-response alert, while reported wellness concerns trigger immediate notification. Notifications distinguish medication misses, no-response events, and wellness concerns, so caregivers can judge urgency without reading the full interaction log, and risk-aware thresholds are applied conservatively to reduce alert fatigue.

Table 2 summarises the three escalation tiers derived from the rule-based risk score computed in Equation (1), where higher weights are assigned to patterns indicating sustained non-response.

$$\text{Risk Score} = 3.0 \cdot \text{CM} + 2.0 \cdot \text{M} + 0.3 \cdot \text{AD} \tag{1}$$

where CM denotes the trailing consecutive-miss streak over all reminders (not per medication), M the number of missed reminders, and AD the average response delay in minutes.

Table 2. Risk score thresholds and corresponding escalation actions.

Risk score	Level	Action
< 5	Low	Normal reminder
5–11	Medium	Follow-up check
≥ 12	High	Alert caregiver

The score has no upper bound; any value ≥ 12 triggers caregiver notification. These thresholds are configurable and intended to be refined through clinical validation in future work.

Access control reflects common eldercare responsibilities. Caregivers authenticate with password-protected accounts, sessions are represented by bearer tokens, and passwords are hashed with bcrypt. The system supports administrator, doctor, and family roles, with per-senior sharing scoped through owner, editor, and viewer access levels. Machine-to-machine endpoints are protected with API keys, and sensitive service credentials are provided through local configuration rather than hard-coded into the application.

5 Evaluation

5.1 Feasibility Tests

We first verified the prototype through scenario-based feasibility tests covering the complete user-facing workflows: medication reminders (scheduling, spoken delivery, delayed follow-up, conversational status extraction, intake-log updates, and escalation after missed-dose outcomes); caregiver-defined wellness check-ins with normal, concerning, and no-response outcomes; everyday current-information questions; and natural-language ad hoc reminder creation. Across repeated scenarios the pipeline executed end to end: status markers were parsed into intake-log updates, missed doses and simulated concern reports (pain, dizziness, loneliness) triggered the configured Telegram and/or SMS notifications, and the text-to-speech fallback chain sustained delivery when the preferred synthesis path was unavailable. These tests establish the breadth of technical feasibility; the pilot below provides depth under real-home conditions.

5.2 In-Home Pilot Setup

To provide initial evidence beyond laboratory feasibility tests, we deployed VAILS for seven consecutive days (8–14 June 2026) in the home of an older adult living independently — a Greek-speaking older woman — on standard consumer hardware. The participant and her family caregiver were informed about the operation of the system and consented to the collection and analysis of the interaction logs; all reported data are anonymised. A family caregiver configured the schedules through the dashboard: five daily medications (a thyroid medication at 08:00, an antihypertensive at 09:30, an anticoagulant at 14:30, a second antihypertensive at 18:45, and a sleep medication at 21:30), one weekly vitamin D supplement, and one daily non-medication reminder.

Reminders were delivered as personalised Greek utterances referencing the appearance of each medication package, with intake confirmed through the voice follow-up dialogue.

5.3 Adherence Results

Over the seven days the system delivered and logged 43 scheduled events: 36 medication reminders and 7 general reminders. Of the 36 medication events, 29 (80.6%) were confirmed as taken through the spoken follow-up, 2 were skipped, and 5 were missed. For confirmed doses, the mean delay between the scheduled time and spoken confirmation was 3.0 minutes (range 2.1–4.3), indicating that the participant responded promptly to the spoken reminders. Table 3 reports compliance per medication. Notably, all non-adherence concentrated in a single medication — the 18:45 antihypertensive, with 0 of 7 doses taken — while every other medication reached 100% compliance. Missed doses triggered the escalation workflow and the corresponding caregiver notifications.

Table 3. Per-medication outcomes over the seven-day pilot (one older adult).

Medication (schedule)	Taken	Skipped	Missed	Compliance
Thyroid medication (daily, 08:00)	7	0	0	100%
Antihypertensive A (daily, 09:30)	7	0	0	100%
Anticoagulant (daily, 14:30)	7	0	0	100%
Antihypertensive B (daily, 18:45)	0	2	5	0%
Sleep medication (daily, 21:30)	7	0	0	100%
Vitamin D (weekly)	1	0	0	100%
Total	29	2	5	80.6%

5.4 ML Analytics on Pilot Data

The per-senior models of Section 4.3 were retrained exclusively on the participant's seven-day pilot log — the synthetic generator served only the feasibility phase, and the rolling feature windows were truncated accordingly — and their outputs were inspected on the caregiver dashboard (Fig. 2). The adherence forecast produced intake probabilities that decrease monotonically across the day — 96.5% at 08:00, 86.4% at 14:00, 70.4% at 18:00, and 53.4% at the 21:00 hour bucket (nearest to the 21:30 sleep-medication slot) — reflecting the direction of the evening miss, though the continuous hour-of-day feature under-predicted this well-adhered dose on this small sample. The risk classifier assigned the participant to the Medium class, consistent with the rule-based risk score of 10.9 computed from the same features (5 missed doses, 3.0 minutes average delay, no consecutive-miss streak across medications), and the dashboard accordingly recommended a follow-up check rather than an urgent alert. On a held-out split of the pilot events the classifier reproduced the rule-derived labels with 100% accuracy — a consistency check, since it is trained and tested on the same near-constant rule labels, not an independent measure of predictive accuracy. The clustering component

(with centres from the accumulated history) assigned the pilot week to the nearest, highest-miss profile, consistent with the concentrated evening misses.

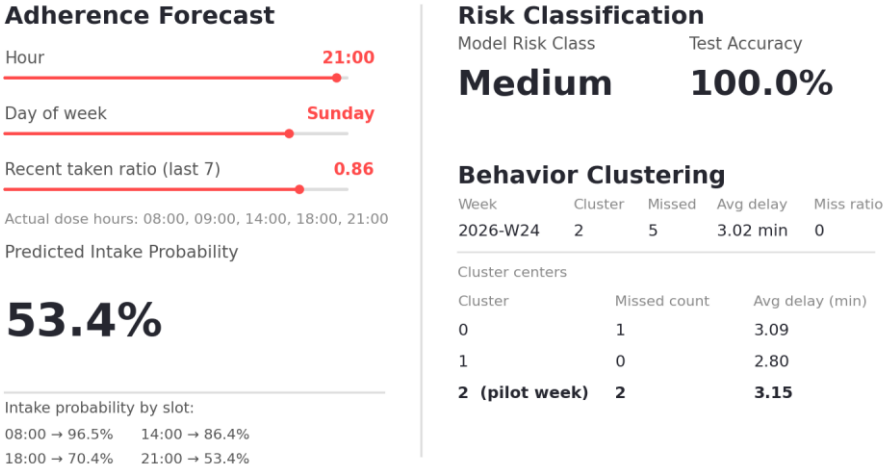


Fig. 2. ML analytics view for the pilot participant at the 21:00 hour bucket (21:30 sleep-medication dose), regenerated from the original dashboard screen to fit the page width.

From the caregiver's perspective, the combined signals converged on one actionable insight: adherence was excellent except for a single evening medication that was never taken. This supported a targeted conversation with the participant and treating physician about that specific dose, rather than a diffuse adherence concern. This illustrates the analytics layer's advisory role: a daily-trend forecast, an urgency level from the risk classifier, and week-over-week context from the clustering, none available from the scalar risk score alone.

This pilot is intentionally modest — a single participant, one home, seven days, no baseline, and self-reported confirmations without pill-count verification. Its value is a first end-to-end demonstration that the complete pipeline operates in a real home with a real older adult, and a template for the larger field evaluation planned as future work.

6 Conclusion and Future Work

We presented VAILS, a voice-first assisted-living system that combines real-time bidirectional speech dialogue, rule-based and ML-driven risk analytics, and a structured caregiver oversight interface into a single cohesive platform. By coupling the Gemini Live API with an energy-based VAD loop, VAILS supports natural spoken reminders and follow-up conversations that require no smartphone literacy from the senior, while the layered TTS chain and configurable escalation protocol improve robustness under varying network and caregiver availability conditions. The ML analytics engine, comprising a logistic regression adherence forecast, a decision tree risk classifier, and k-means behavioural clustering, transforms intake logs into interpretable caregiver-facing

signals without requiring pre-labelled datasets or domain expertise to configure. A preliminary seven-day in-home pilot with one older adult provided first evidence that the complete pipeline operates end to end in a real home and that the analytics translate raw logs into targeted caregiver guidance.

Together, these components address the core limitations identified in Section 3: inaccessible voice-first interaction, one-way notification without intake confirmation, and simple threshold alerts rather than personalised risk modelling. The conversational layer extends VAILS toward a broader daily-living assistant, and demonstrates that a lightweight stack — FastAPI, Streamlit, APScheduler, scikit-learn, and the Gemini Live API — can support a functional assisted-living prototype suitable for future field evaluation.

6.1 Limitations

The current VAD is energy-based (RMS threshold = 250) and may generate false positives in noisy domestic environments. The ML models are trained on individual intake histories and may require several weeks of data before producing reliable predictions for a new senior. Privacy considerations regarding always-on microphone access must be addressed before broad deployment, including transparent data governance and local audio processing options. The system currently targets Greek-speaking environments; multilingual support and cultural adaptation remain outstanding engineering challenges. Finally, the functional tests were conducted on a single development device with simulated interactions, and the in-home pilot involved a single participant over seven days without a baseline or a directly comparable, publicly evaluated system. Usability, acceptability, and long-term adherence outcomes can therefore only be established through future clinical studies with target populations.

6.2 Future Work

Several directions are planned for the next development phase, with a primary focus on enhancing medication adherence support while extending the system's broader assisted-living capabilities.

First, building on the preliminary pilot reported here, we plan a larger field evaluation with older adults and caregivers to assess usability, acceptability, adherence-support workflows, and caregiver workload, and to collect the high-quality multi-user dataset needed to validate the ML models against clinical ground truth. Further technical directions include a learned speech/non-speech classifier for noise-robust voice activation, sequence-aware models that capture temporal dependencies in user behaviour, integration with national electronic prescription systems to pre-fill medication schedules, and evaluation of the dialogue module with older adults with mild cognitive impairment.

Planned extensions include multi-user profiles within a single deployment with robust user identification, multi-room coverage coordinating several audio-only devices, voice-mediated teleconsultation with family members or physicians, and light cognitive stimulation activities that support engagement while providing additional longitudinal

behavioural signals. Privacy-preserving aggregation of locally trained models across users could improve shared ML models without direct sharing of personal data.

Before deployment, regulatory compliance will also need to be addressed. VAILS processes health-related personal data and employs AI-driven decision support, placing it within the scope of the GDPR and potentially the EU AI Act, which classifies AI systems used in healthcare contexts as high-risk. A formal compliance assessment will therefore be part of future work.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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